Alex Meredith
Professor Doluweera
PHYS 3550

Computational Studies on the Physical Pendulum

Introduction

Two different formulas were used to calculate the moment of inertia of the pendulum with and without a wooden block attached. The Huygens–Steiner theorem details the methodology used to find the moment of inertia without a wooden block and with the pendulum being solely comprised of a meterstick, as the pivot point was a given distance from the center of the meterstick. For trials where the wooden block was part of the pendulum, the moment of inertia of the block, found by multiplying the block's mass by the squared value of the block's distance from the pendulum's pivot, is added to the Huygens–Steiner moment of inertia. For small angles, $\sin\theta \approx \theta$, meaning that this substitution applies for equations of damped pendulums without drag, with linear drag, and with quadratic drag.

$$I\,\frac{d^2\theta}{dt^2} + mgl\sin\theta = 0$$

$$I\,\frac{d^2\theta}{dt^2} + b\frac{d\theta}{dt} + mgl\sin\theta = 0$$

$$I\,\frac{d^2\theta}{dt^2} + k|\omega|\frac{d\theta}{dt} + mgl\sin\theta = 0$$

$$I = I_c + md^2$$

$$I = I_c + md^2 + m_b d_b^2$$

Method

The linear damping coefficient b and quadratic drag coefficient k were isolated from their respective differential equations and solved for using equations 6 and 7. The least squares method was used to find a single value of b and k, as using equations 6 and 7 would result in arrays being printed in the Python code. To reduce noise, data values where the absolute value of v was close to or near zero were omitted, as otherwise there would be incorrect spikes that altered the output for both constants. In order to find realistic values for the

linear damping coefficient and quadratic drag, the Savitzky–Golay filter was applied to the data for run #3, which greatly improved the accuracy of the returned values.

$$b = -\frac{I\alpha + mgl\sin\theta}{\omega}$$

$$k = -\frac{I\alpha + mgl\sin\theta}{\omega|\omega|}$$

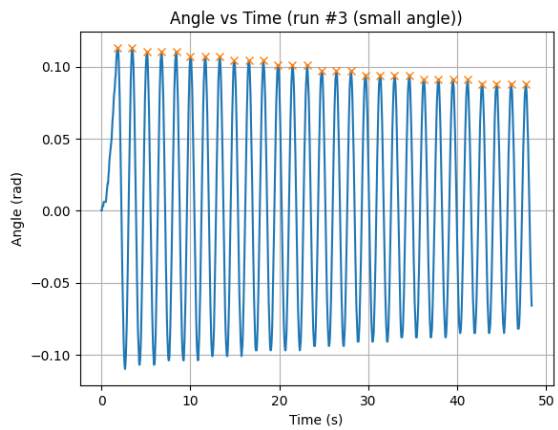
$$\alpha = -\frac{b\omega}{I} - \frac{mgl\sin\theta}{I}$$

$$T = \frac{2\pi}{\sqrt{\frac{mgl}{I}}}$$

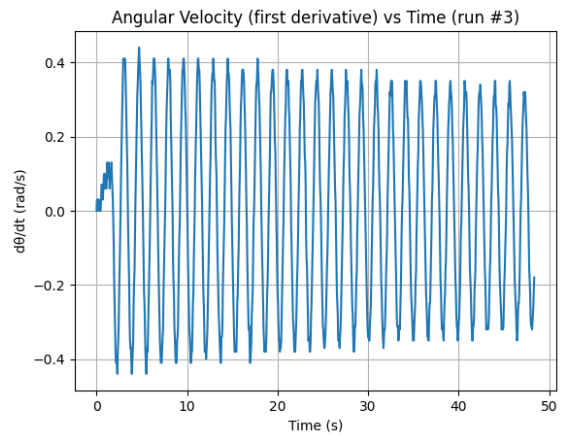
Results

For runs #3 and #4, where the pivot point is 49 cm from the center of the pendulum, the moment of inertia was found to be $0.0388\,\mathrm{kg\,m^2}$. The moment of inertia of the meterstick itself was found to be $0.010\,\mathrm{kg\,m^2}$. The calculated angular velocity matches the values and the behavior seen in the given data, as the data has its first peak at 0.419 rad/s, coinciding with the peak from the graph. In run #4, the peak angle value was 0.465 radians, while the peak angular velocity was around 1.7 rad/s, which agrees with the calculated values seen in the graph. For runs #19 and #20, the maximum angle and angular velocity values of 0.072 radians, 0.272 rad/s, 0.427 radians, and 1.634 rad/s agree with the calculations seen in their respective graphs. For runs 21 and 22, the peak values from the given data are 0.126 radians, 0.503 rad/s, 0.487 radians, and 1.843 rad/s, which correspond visually to their respective graphs. Using equation 9, the theoretical small-angle period for run #3 was determined to be 1.14 seconds. After using the Savitzky–Golay filter to filter out noise and subsequently smooth the data points, the numerical value for the linear damping coefficient was found to be 0.0314, while the quadratic drag coefficient k was found to be 0.003391. When the pendulum was comprised of a wooden block attached to a meterstick, the moment of inertia of each body was found separately, then added together for a total of $0.3329\,\mathrm{kg\,m^2}$. The graph of the undamped Euler–Cromer simulation has no decay in its absolute value, unlike the other two functions with coefficients. This behavior is expected and is evidence of a positive trial, as without any resistance force, the pendulum is expected to harmonically oscillate eternally.

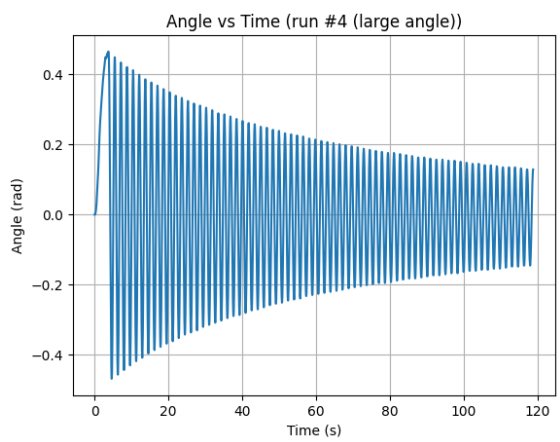
Graphs for Runs #3 and #4



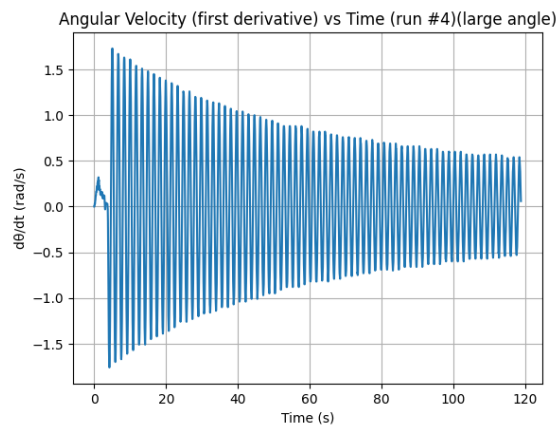
(a) Run 3: $\theta(t)$



(b) Run 3: $\omega(t)$



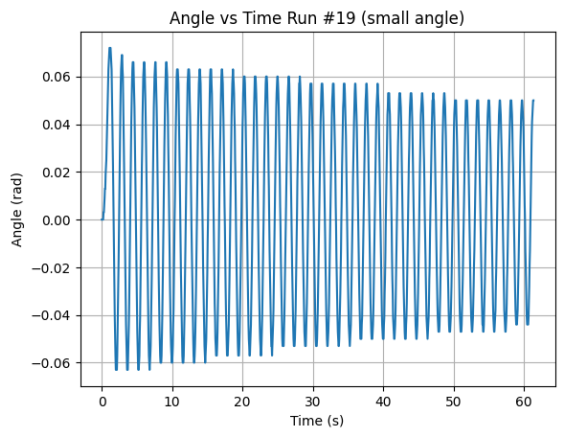
(c) Run 4: $\theta(t)$



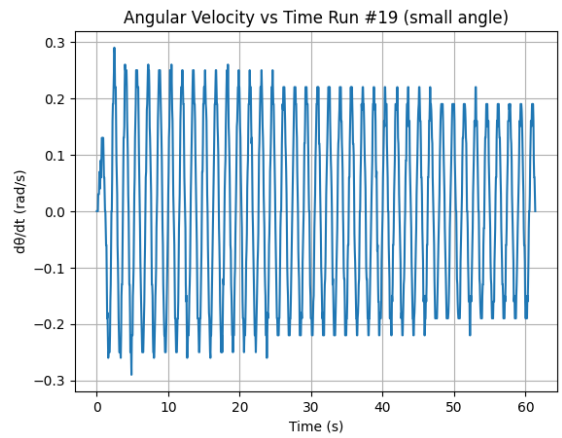
(d) Run 4: $\omega(t)$

Figure 1: Angular position and angular velocity for Runs #3 and #4.

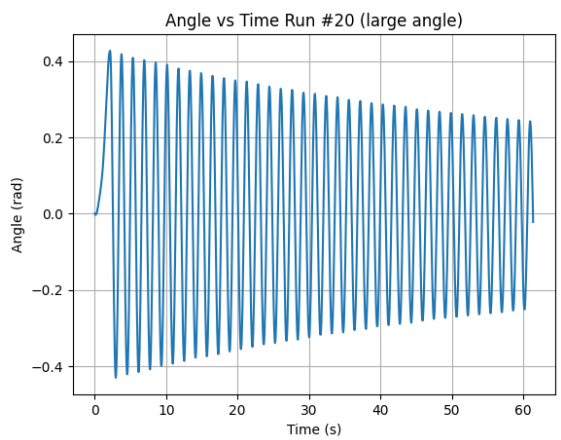
Graphs for Runs #19–21



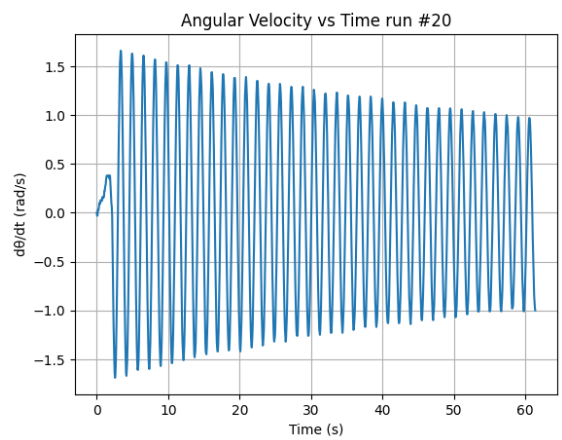
(a) Run 19: $\theta(t)$



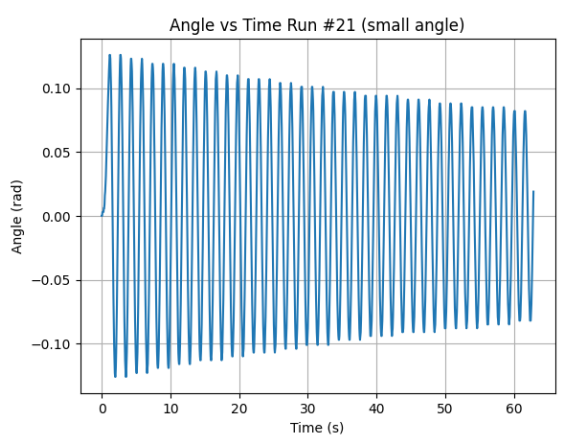
(b) Run 19: $\omega(t)$



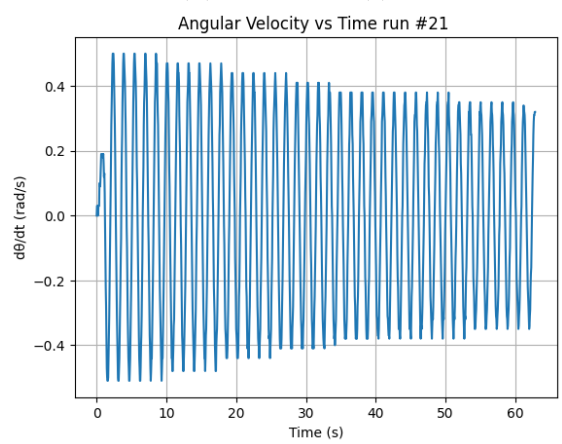
(c) Run 20: $\theta(t)$



(d) Run 20: $\omega(t)$



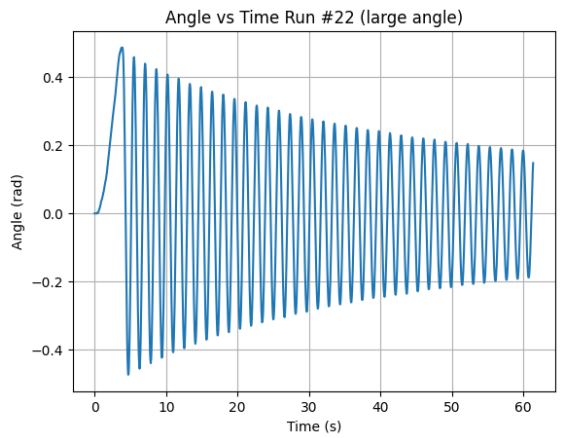
(e) Run 21: $\theta(t)$



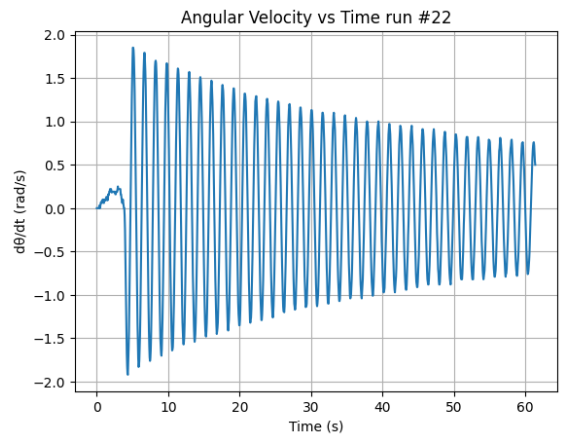
(f) Run 21: $\omega(t)$

Figure 2: Angular position and angular velocity for Runs #19–21.

Graphs for Run #22



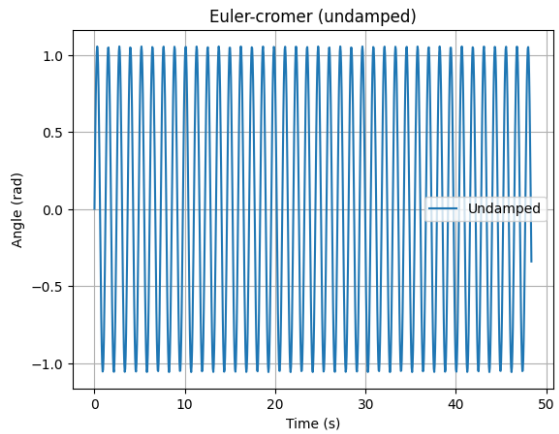
(a) Run 22: $\theta(t)$



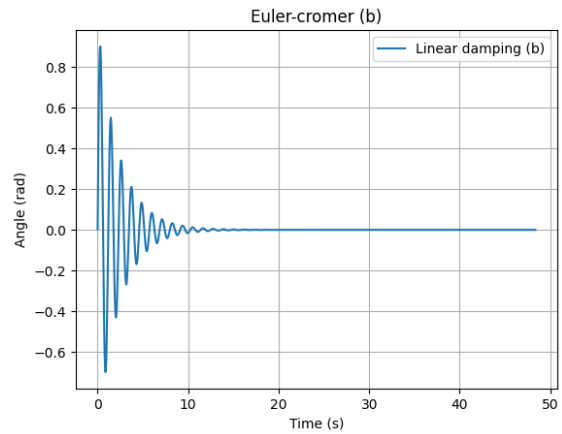
(b) Run 22: $\omega(t)$

Figure 3: Angular position and angular velocity for Run #22.

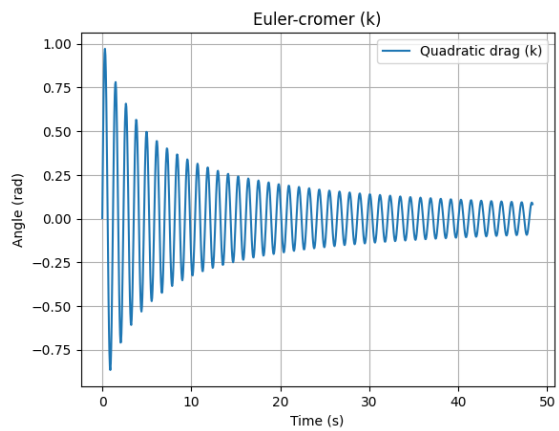
Euler–Cromer Method Simulations



(a) Undamped



(b) Linear damping



(c) Quadratic drag

Figure 4: Euler–Cromer simulations of undamped, linearly damped, and quadratically damped pendulum motion.

Discussion

Period and frequency are closely related to a pendulum's moment of inertia and the distance from its center of mass to its pivot position. In runs #3 and #4, the pivot position was 49 cm from its center of mass, and the moment of inertia was found to be 2.622 times greater than the same pendulum with a pivot position only 20 cm from its center. As seen in equation 9, a larger moment of inertia results in a longer period, which can be seen theoretically as a slower oscillation. This agrees with the experimental angular velocity data, as the angular velocity from run #3 is greater than 0.4 rad/s, while the angular velocity from run #19 is less than 0.3 rad/s. Larger initial amplitudes can slightly increase the period, while the small-angle approximation assumes $\sin\theta \approx \theta$. Air drag reduces the amplitudes as time increases,

which can be represented numerically by the linear damping and quadratic drag coefficients, b and k. One interesting idea that could be done to slightly change the project could be to compare the resulting data from simply releasing the pendulum vs. applying a push to the pendulum.